Blood Pressure Goal: 130/80 mmHg Vs 140/90 mmHg

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Clinical Pharmacist

Learning Objectives

- Review current guideline recommendations regarding blood pressure goals.
- Evaluate clinical trials that studied outcomes in patients with different blood pressure targets.
- Create a plan for establishing a blood pressure goal based on patient-specific factors.

Patient Case Scenario

- "You are an ambulatory care pharmacist serving chronic disease state management for a facility with a large population of patients with diabetes. A male patient that has been under your care notes confusion regarding his blood pressure goal. The last time he saw you, he was told to target a blood pressure goal of **140/90 mmHg**. Recently, he visited his PCP and reports being told that anything **below 130/80 mmHg** was fine. He is back for a return visit with you, and wants to know what his "real" blood pressure goal is to prevent further confusion. He is rather frustrated and feels that there should just be one goal. Please share how you would respond to the patient, including your rationale."

Background

- Blood pressure (BP) is the pressure of circulating blood on the walls of blood vessels.[1]

- Hypertension (HTN) is a long-term medical condition in which the blood pressure in the arteries is persistently elevated.[2]

- Long-term HTN puts patients at greater risk for heart disease, stroke, kidney disease, and other complications.[1]

1. The Lancet. 389 (10064): 37–55. doi:10.1016/S0140-6736(16)31919-5. PMC 5220163
2. Naish J, Court DS (2014). Medical sciences (2 ed.). p. 562. ISBN 9780702052491.

HTN Classifications

Guideline	JNC 8 (2014)		AHA/ACC (2017)	
Category	Systolic BP (mmHg)	Diastolic BP (mmHg)	Systolic BP (mmHg)	Diastolic BP (mmHg)
Normal	<120	<80	<120	<80
Pre-hypertension/Elevated	120-139	80-89	120-129	<80
Stage 1	140-159	90-99	130-139	80-89
Stage 2	>160	>100	>140	>90

1. "Eighth Joint National Committee (JNC 8), "2014 Evidence-based Guideline for the Management of High Blood Pressure in Adults," December 2013.
2. "2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults," November 2017.

Notable Clinical Trials

- ACCORD BP (2010)
- SPRINT (2015)

ACCORD BP (2010)

- Clinical Question:
  - In patients with **T2DM at high risk for CV events**, does intensive BP control (SBP <120 mmHg) reduce rates of nonfatal MI, nonfatal stroke, or CV mortality when compared to standard BP control (SBP <140 mmHg)?

NEJM. 2010. 362(17):1575-1585.

ACCORD BP (2010)

Overview	Primary Outcome	Inclusion Criteria

Multicenter, randomized, controlled, open-label trial
 	
77 Centers in Canada and US
 	
Open-label randomization to a group:
- Intensive BP Control – Goal SBP < 120 mmHg (n=2,362)
- Standard BP Control – Goal SBP < 140 mmHg (n=2,371)
 	
Median Follow-up: 4.7 years
	First occurrence of composite of non-fatal MI, non-fatal stroke, or CV death.	• Type 2 diabetes mellitus
		• Hemoglobin A1C ≥ 7.5%
		• Age ≥ 40 years with CVD
		• Age ≥ 55 years with any of the following:
		  • Atherosclerosis
		  • Albuminuria
		  • Left ventricular hypertrophy
		  • ≥ 2 CV risk factors*
		  *(dyslipidemia, hypertension, smoking, or obesity)

Outcome and Results

- **Primary outcome**: Nonfatal MI, nonfatal stroke, or CV mortality: 1.87%/yr vs. 2.09%/yr (HR 0.88; 95% CI 0.73-1.06; P=0.20)
- **Conclusion:** With a median follow-up of 4.7 years, there was no difference in nonfatal MI, nonfatal stroke, or CV mortality between the two groups.

NEJM. 2010. 362(17):1575-1585.

SPRINT (2015)

- Clinical Question:
  - In patients at **high risk for CVD** but who **do not have a history of stroke or diabetes**, does intensive BP control (target SBP <120 mm Hg) yield superior CV outcomes compared to standard treatment (target SBP 135-139 mm Hg)?

SPRINT (2015)

Overview	Primary Outcome	Inclusion Criteria

Multicenter, open-label, randomized controlled trial
 	
102 sites in US and Puerto Rico
 	
9,361 patients randomized
 	
Open-label randomization to a group:
- Intensive BP Control – Goal SBP < 120 mmHg (n=4,678)
- Standard BP Control – Goal SBP 135-139 mmHg (n=4,683)
 	
Median Follow-up: 3.26 years (stopped early; planned 5 years)
	First occurrence of MI, ACS, stroke, HF, or CV mortality	• Age ≥ 50 years
		• SBP: 130-180 mm Hg (treated or untreated)
		• Additional cardiovascular disease (CVD) risk: (at least one)
		  • Clinical or subclinical CVD (excluding stroke)
		  • Chronic Kidney Disease (CKD), defined as eGFR 20 – 59 ml/min/1.73m^2 or MDRD in prior 6 months
		  • Framingham Risk Score for 10-year CVD risk ≥ 15% in prior 12 months
		  • Age ≥ 75 years

Outcome and Results
• **Primary outcome**: First ACS, stroke, HF, or CV death: 5.2% vs. 6.8% (HR 0.75; 95% CI 0.64-0.89; P<0.001; NNT 63) • **Conclusion:** With a median follow-up of 3.26 years, intensive blood pressure control was associated with fewer fatal or non-fatal cardiovascular events.

NEJM. 2015. 373(22):2103-2116.

Blood Pressure Goals

JNC 8 (2014)	AHA/ACC (2017)	ADA (2020)
• BP <140/90 mmHg: age 18-59 years (no DM or CKD) • BP < 140/90 mmHg: CKD and/or DM (all ages) • BP < 150/90: age ≥ 60 years old (no DM or CKD)	BP <130/80 mmHg	• BP <140/90 mmHg: DM, HTN, and 10-year ASCVD risk < 15% • BP <130/80 mmHg: DM, HTN, and 10-year ASCVD risk ≥ 15%* *stricter goal may be appropriate, if it can be tolerated

1. "Eighth Joint National Committee (JNC 8), "2014 Evidence-based Guideline for the Management of High Blood Pressure in Adults," December 2013.
2. "2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults," November 2017.
3. American Diabetes Association. Standards of Medical Care in Diabetes—2020. Diabetes Care. 2020;43(suppl 1):S1-S212.

Back To The Patient

Patient Case Scenario

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  - Some guidelines recommend stricter BP goals than others, due to CV risk.
  - There are some studies showing benefit in stricter goals, however there is still some debate.
  - 130/80 mmHg may be a suitable goal, if tolerated.

Conclusion

- Guidelines can differ and are not a substitute for clinical judgement.
- Goals should be based on patient-specific factors.
- It is important to educate and support the patient when reaching these goals.

Treat Patients, Not Numbers!

Image: http://www.clker.com/cliparts/f/2/d/b/15164167921513496842doctor-checking-patient-clipart.med.png

Questions?

- Feel free to contact me with any questions:

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